Experiment 7

Q Cheapest flights with K Stops

Ans: Approach: Shortest path problem with constraint (K stops)

→ Can't be done using normal dijkstra, track it don't number of stops

→ We need to calculate
- Cost so far
- Stops used

Pseudocode:

```
int findCP(int n, vector<vector<int>> &flights,
int src, int destination, int k){

  vector<pair<int, int>> adj[n];

  for (auto &f : flights){
      adj[f(0)].push_back({f(1), f(2)}); }

queue<pair<int, int>> q;
  q.push({src, 0});

vector<int> dist(n, INT_MAX);

dist[src] = 0;
int stops = 0;
while(!q.empty() && stops <= k){

  int size = q.size();
```

```
while (size--) {
    auto [node, cost] = q.front();
        q.pop();
    for (auto & it : adj[node]) {
        int next = it.first;
        int price = it.second;
        if (cost + price < dist[next]) {
            dist[next] = cost + price;
            q.push({next, dist[next]});
        }}
    stops ++;
}
return dist[dst] == INT_MAX ? -1 :
    dist[dst];
}
```

Complexity: Space = $O(k * 6)$

time = $O(N+6)$